



Improved latitudinal carbon budgets from global airborne surveys

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Robust information on the spatial distribution of global carbon fluxes is required to project the future trajectory of carbon-climate feedback effects and atmospheric CO₂ concentrations. Estimates of the latitudinal partitioning of carbon fluxes from top-down atmospheric CO₂ inverse models currently diverge widely, because of methodological limitations or systematic biases in models or observations. We use airborne CO₂ observations from the NASA Atmospheric Tomography Mission to evaluate and refine inverse model estimates from the Orbiting Carbon Observatory version 10 Model Intercomparison Project of total CO₂ exchange for the two-year period of June 2016–May 2018. Applying emergent concentration-flux relationships as constraints reduces zonal total flux uncertainties by 46 to 56% relative to the full v10 MIP ensemble and by 17 to 28% relative to the subset excluding satellite observations over ocean. Subtracting independent estimates of fossil-fuel emissions and air-sea gas exchange results in residual land fluxes with a large northern extratropical sink, a small southern extratropical sink, and a small tropical source. The airborne-derived tropical land source disagrees with a large tropical land sink from process-based terrestrial models combined with estimates of land use emissions and river fluxes, representing an important challenge for our understanding of the global carbon cycle. The large implied northern extratropical sink can be explained either by underestimated land uptake by process models or a combination of process model bias and overestimated fossil fuel emissions.

global carbon cycle | airborne observations | satellite CO₂ measurements | atmospheric inverse modeling

Global carbon cycle science is at a critical juncture, where society's need for accurate information on the anthropogenic and natural emissions and removals of CO₂ to and from the atmosphere is rapidly outpacing the scientific community's ability to provide it. Robust estimates of the large scale sources and sinks of CO₂ are required for better understanding and prediction of carbon-climate feedback effects, which are an important contributor to uncertainty in global climate projections (1, 2). Of these, the response of the terrestrial biosphere to increasing atmospheric CO₂ and shifting temperature, precipitation, nutrient, and disturbance patterns contributes the most uncertainty (2). The relative magnitudes of these effects manifest in the present-day latitudinal distribution of net land carbon uptake. For example, if the land sink is predominantly in the tropics, a strong CO₂ fertilization effect would be the most likely cause (e.g., refs. 2–4). Conversely, a stronger extratropical sink would suggest a bigger role for regrowth following past disturbance or positive climate or nutrient effects on plant growth at high latitudes, or both (e.g., refs. 5–8). There is presently a large discrepancy between Earth system models, which have strong fertilization effects and consequently stronger tropical land carbon uptake (1), and inverse model ensembles which estimate greater extratropical uptake (8, 9). Thus, improved observational constraints on the latitudinal distribution of carbon sinks will provide greater clarity for process model improvements and more accurate climate projections.

It is generally not possible to integrate local scale CO₂ flux measurements to quantify CO₂ emissions and uptake at regional to global scales, owing to high spatial heterogeneity and unmeasured diffuse natural fluxes. However, the atmosphere integrates flux signals from many heterogeneous natural and industrial processes. Thus, the inversion of atmospheric CO₂ concentration measurements using models of atmospheric transport has long held promise for quantifying regional to global-scale surface CO₂ exchange (e.g., refs. 10–12). Despite considerable effort by both the measurement and modeling

Significance

We show that large scale CO₂ patterns measured by global airborne surveys are able to discriminate among a collection of diverse atmospheric models that estimate total CO₂ fluxes. In combination with independent ocean and fossil-fuel flux estimates, the aircraft observations indicate much less tropical land uptake than suggested by terrestrial ecosystem models and land use studies, with important implications for the processes driving the global land carbon sink. In the northern extratropics, the observations show either an underestimate of land carbon uptake by models or an overestimate of emissions by fossil fuel inventories. Our results highlight the unique value of global-scale airborne atmospheric profile measurements for the future global carbon observing system.

The authors declare no competing interest.

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communities, large-scale zonal flux estimates have not converged with each other (8, 9, 13, 14), as a result of persistent technical challenges. Historically, these challenges have included atmospheric transport model biases, surface network sparseness, and the mathematically ill-posed nature of the inverse problem for CO₂ flux estimation. Since the advent of satellite CO₂ measurements, these challenges now also include systematic measurement biases and poor satellite coverage at high latitudes and in cloudy regions.

The carbon-cycle research community now has many decades of surface CO₂ concentration measurements from a network of over 100 global stations, and over 10 y of satellite total column average CO₂ measurements with near global coverage. In contrast, in situ CO₂ observations remain sparse above the surface. However, high-resolution vertical profiles, such as those provided by airborne measurements, have distinct advantages over surface observations because they integrate flux signals over much larger space and time scales (e.g., refs. 15 and 16), and averaging profile observations within and above the boundary layer can largely overcome model biases in vertical mixing (e.g., refs. 17 and 18). Other advantages over both surface and satellite observations include that airborne observations can provide independent constraints on atmospheric transport biases in models (e.g., refs. 16, 19, and 20), and that using a single instrument or set of instruments over large distances can provide much more precise estimates of spatial gradients in CO₂ concentration (e.g., refs. 15 and 16).

Here we use airborne observations from the NASA Atmospheric Tomography Mission (ATom, 21), which included coincident CO₂ measurements from five instruments—three in situ and two whole air samplers to allow for rigorous quantification of measurement error—spanning from the High Arctic to Antarctica and the surface to the lower stratosphere, over four deployments in different seasons during 2016–2018 (*Materials and Methods*). We also use coincident output from the NASA Orbiting Carbon Observatory (OCO-2) version 10 inversion model intercomparison project (MIP) (9) (*Materials*

and Methods). The OCO science team and the Global Carbon Project have conducted a series of inversion MIPs that highlight the evolving state of global atmospheric CO₂ inversion uncertainty (8, 9, 14, 22–24). The OCO-2 v10 MIP included five experiments using different combinations of in situ and satellite observations: in situ, mostly surface, observations only (the IS experiment); satellite observations over ocean in glint viewing mode only (the OG experiment); satellite observations over land in nadir or glint viewing mode (the LNLG experiment); satellite land nadir and glint, and in situ observations combined (the LNLGIS experiment), and all satellite and in situ data combined (the LNLGOGIS experiment). Here we use the 10 OCO-2 v10 MIP models which output ATom cosamples for each of the five experiments, for a total of 50 inversions. We use the airborne observations and cosampled model output to calculate observed and modeled pressure-latitude cross-sections of annual-mean CO₂ concentration. We identify a dominant pattern of variability across inversions corresponding to higher or lower CO₂ concentration in the tropics versus extratropics, and strong correlations between this pattern and total CO₂ fluxes estimated for broad latitudinal zones. We then use these emergent concentration-flux relationships to identify biases in and narrow the uncertainty of the ensemble total flux estimates of the OCO-2 v10 MIP inversions. Finally, we incorporate independent estimates of fossil-fuel and ocean CO₂ fluxes to provide residual land flux estimates and compare these to a combination of process-based dynamic global vegetation models (DGVMs) and estimates of land-use change emissions and river transport.

Results and Discussion

Flux Dipoles. Fig. 1 highlights several prominent flux dipoles in the OCO-2 v10 MIP results. Of first-order importance, atmospheric inversions are unable to adequately distinguish global land and ocean fluxes (Fig. 1A); the models with small ocean-versus-land spread across experiments (e.g. CMS-Flux,

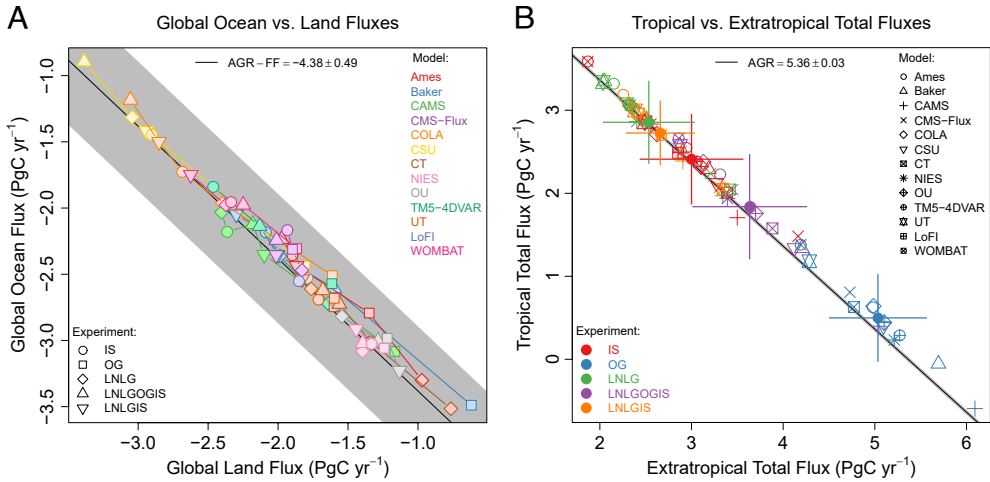


Fig. 1. Global-scale dipoles in OCO-2 v10 MIP posterior fluxes (positive to the atmosphere) for the full 2015–2020 intercomparison period. (A) global ocean versus global land fluxes colored by model with symbols indicating experiment. The black line shows the difference between the NOAA atmospheric growth rate (AGR, 25) and fossil-fuel emissions (FF, 8) which should equal the sum of global land and ocean fluxes by mass balance. The gray shading shows 1- σ uncertainty on 2015–2020 mean AGR-FF, which is dominated by the reported uncertainty on the fossil-fuel estimate. (B) tropical (20°S–20°N) total (ocean, land, and fossil) flux versus extratropical (poleward of 20° in either hemisphere) total flux colored by experiment with open symbols indicating model. Solid circles and whiskers show experiment means and SDs. The black line shows the NOAA AGR, and the gray shading shows 1- σ uncertainty in the NOAA AGR over this period. The AGR and FF values reported here differ from the sum of those in Table 1 owing to the differing time periods. The tight correlations here are forced by mass balance, and also in panel (A) by the use of a single common fossil fuel estimate in the inversions. Of more interest is the magnitude of spread, and relationships between models and experiments, in the $-1:1$ direction. SI Appendix, Fig. S1 shows the same points but with the coloring by model or experiment in each panel reversed.

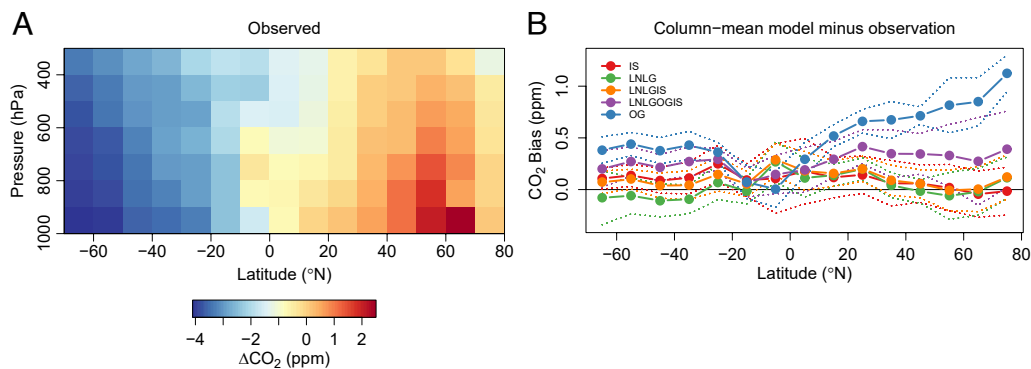


Fig. 2. Observed annual-mean CO_2 distribution and partial-column annual-mean inversion offsets over the period June 2016–May 2018. (A) Pressure-latitude cross-section from harmonic fits to detrended and bin-averaged ATom observations. Zero corresponds to the annual-mean at MLO, which is subtracted in the detrending. (B) 10-degree binned differences between cosampled inversion output and corresponding observed pressure-weighted column means (1,000 to 300 hPa) averaged over all models within each experiment. Thin dotted lines show the $\pm 1\text{-}\sigma$ model spread within each experiment.

UT, or NIES) achieve this by enforcing tight adherence to their a priori ocean fluxes, while those with looser ocean prior constraints differ by up to 1.5 PgC y^{-1} across experiments (*SI Appendix, Fig. S2*). Furthermore, prior fossil-fuel emissions in these inversions are assigned zero uncertainty, but contribute an additional $\pm 0.4 \text{ PgC y}^{-1}$ ($1\text{-}\sigma$) uncertainty (8) to the inferred total ocean plus land sink (*Fig. 14*). Recently, Randerson et al. (26) suggested that global fossil fuel emissions might be biased high by 6% ($\sim 0.6 \text{ PgC y}^{-1}$), with important implications for the size and location of the inferred global land sink. Here, to avoid ocean versus land trade-offs or aliasing fossil-fuel emission biases, we evaluate the total surface CO_2 exchange comprised by the sum of ocean, land, and fossil fluxes, as seen by the atmosphere.

Fig. 1B shows that the OCO-2 v10 MIP models also differ substantially in their estimates of tropical versus extratropical total CO_2 fluxes. On this axis of variability, the choice of input data across experiments has a primary influence, with inversions including satellite ocean glint observations (LNLGOGIS and especially OG) producing much larger extratropical and much weaker tropical CO_2 emissions than those without. This flux dipole is particularly important for the understanding of the processes responsible for net land carbon uptake and potential future carbon-climate feedback.

Mean Concentration Patterns. The ATom-observed bin-averaged (*Materials and Methods*) annual-mean CO_2 cross-section is shown in *Fig. 2*, as well as the partial column mean inversion-minus-observation differences averaging across models within each experiment. The overall annual-mean pattern reflects the strong fossil-fuel CO_2 emissions at northern mid-latitudes, moderated by atmospheric mixing and exchanges with the ocean and terrestrial biosphere. On average, the LNLGOGIS and OG experiments overestimate CO_2 in the northern extratropics by 0.3 and 0.8 ppm, respectively, and in the southern extratropics by 0.2 and 0.4 ppm, respectively. IS, LNLG, and LNLGIS biases are smaller, at +0.1 to +0.25 ppm in the tropics, and +0.05 to 0.1 ppm (IS and LNLGIS) or -0.1 ppm (LNLG) in the southern extratropics. We further evaluate variability across inversions with the pressure-latitude bin-average cross-inversion SD and first empirical orthogonal function (EOF1), as shown in *SI Appendix, Fig. S3* and discussed in *SI Appendix, Text*.

Concentration-Flux Correlations. To explore the relationships between broad meridional CO_2 concentration patterns and fluxes from specific zones, we have calculated correlations across

inversions between their annual-mean posterior concentrations for each pressure-latitude bin and their respective annual total fluxes within three zonal bands: southern extratropics (SET, $>20^\circ\text{S}$), tropics (T, 20°S – 20°N), and northern extratropics (NET, $>20^\circ\text{N}$). The resulting correlation coefficients are shown in *Fig. 3*. These correlation patterns are remarkably coherent, with extratropical fluxes positively correlated with extratropical concentrations and negatively correlated with tropical concentrations (*Fig. 3A and C*), and vice versa for the tropics (*Fig. 3B*). The very similar correlation patterns across *Fig. 3A–C*, and the positive correlations between extratropical fluxes and concentrations in the opposite hemisphere, result because of correlations between the underlying zonal fluxes. The pattern of positive correlation between northern extratropical and southern extratropical fluxes, and anticorrelation of both to tropical fluxes, is likely related to the OCO-2 v10 empirical bias correction, which is strongly latitudinally dependent and symmetric about the Equator, with approximately 2 ppm differences between the tropics and extratropics (27). The near-field extratropical correlations are somewhat stronger, for example NET concentrations versus NET fluxes and SET concentrations versus SET fluxes, as expected. For all three flux zones, the correlations to concentrations in the tropics are centered south of the Equator and strongest from 10 to 0°S . The flux correlations in extratropical regions appear weaker near the surface (*Fig. 3*). These diminished correlations may result from transport model differences in boundary-layer mixing that operate independently of the stronger, more consistent effects arising from variations in input data and output fluxes.

Empirical Concentration-Flux Relationships. We seek to quantitatively relate concentration patterns and fluxes across inversions. A simple way to do this might be to calculate latitudinal gradients by differencing concentrations averaged over extratropical and tropical regions, similar to the surface station differences used in Ciais et al. (28). However, this approach would equally weight cross-inversion variations in the two regions, whereas the correlation patterns in *Fig. 3* show that the strength of the concentration-flux relationships vary with region. Instead, we use multiple linear regression (MLR) to model the relationship between fluxes and concentrations averaged over several broad extratropical and tropical regions.

We use the ATom cosampled inversion concentrations to predict the inversion fluxes according to:

$$F_{\text{NET}} = m_T \times C_T + m_N \times C_{\text{NET}} + b \quad [1]$$

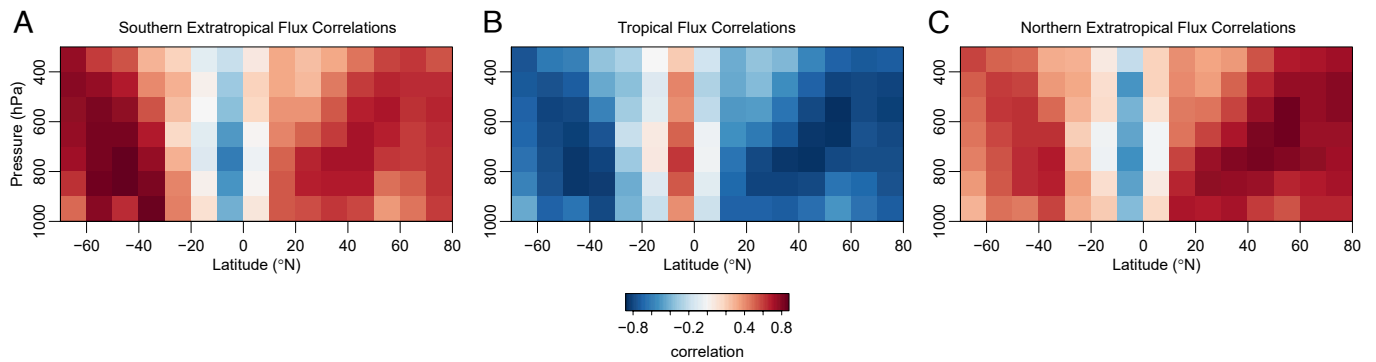


Fig. 3. Zonal concentration-flux correlation patterns in v10 MIP. Across-inversion ($n = 50$) correlation coefficients between annual-mean concentration at each pressure-latitude bin and annual mean total CO_2 fluxes for three zones: the (A) southern extratropics ($>20^\circ\text{S}$), (B) tropics (20°S – 20°N), and (C) northern extratropics ($>20^\circ\text{N}$). Annual-mean concentrations for each inversion and bin come from harmonic fits to bin-averaged and detrended ATom cosamples as described in the text. Fluxes and concentrations are from inversion output June 2016–May 2018.

$$F_T = m_S \times C_{SET} + m_T \times C_T + m_N \times C_{NET} + b \quad [2]$$

$$F_{SET} = m_S \times C_{SET} + m_T \times C_T + b, \quad [3]$$

where the coefficients m_S , m_T , m_N , and b take different values in each equation and the flux (F) and concentration (C) terms are defined as follows. We divide the fluxes at 20° in either hemisphere, but on the basis of the correlation patterns evident in Fig. 3, we use slightly shifted cutoffs for averaging southern extratropical and tropical concentrations. To predict northern extratropical average flux (F_{NET} , $>20^\circ\text{N}$), we use northern extratropical average concentration (C_{NET} , 20°N – 80°N) and tropical average concentration (C_T , 10°S –Equator). To predict southern extratropical average flux (F_{SET} , $>20^\circ\text{S}$), we use southern extratropical average concentration (C_{SET} , 30°S – 70°S) and C_T . To compare to tropical average flux (F_T , 20°S – 20°N), we use C_T , C_{NET} , and C_{SET} in a three-variable MLR, with all three variables having statistical significance following model fitting. For both F_{NET} and F_{SET} , adding a third variable, for example C_{SET} for F_{NET} or vice versa, does not explain additional variance, so we exclude these terms. In each case, we average the concentrations in each region between 1,000 to 300 hPa with cosine weighting to account for decreasing zonal volumes at higher latitudes. The resulting MLR coefficients and their SEs from fitting [1]–[3] to the 50 inversions (10 models for each of five experiments) are shown in SI Appendix, Table S2.

The resulting correlations between inversion fluxes and fluxes predicted by the MLR fits are shown in Fig. 4. The MLR correlations are strong, with adjusted r^2 values of 0.79, 0.84, and 0.71 for F_{SET} , F_T , and F_{NET} , respectively. Consistent with the tropical versus extratropical dipole illustrated in Fig. 1B, these correlations are driven primarily by the changing input observations across experiments. For the experiment means, which average over all models run with the same selection of input data, all r^2 values are greater than or equal to 0.96.

Improved Total Flux Estimates. Emergent constraints (ECs) are a common method for evaluating climate projections, in which a correlation is found across an ensemble of Earth system models (ESMs) between a future property of the system and an observable feature of the present day or historical climate, with corresponding observations used to refine the collective model estimate (e.g., refs. 29–32). The same statistical approach has also been applied to constrain aspects of the present day global carbon cycle, by correlating non-directly-observable fluxes and observable concentration variations across ensembles of

atmospheric transport models (e.g., refs. 15 and 33–35). We refer to our carbon cycle application as an emergent concentration-flux constraint (ECFC). As applied here, our method also differs from a traditional EC in that we allow for more than one predictor variable. Specifically, we use the correlations between zonal-mean total fluxes and tropospheric column-mean ATom cosample concentrations for two or three broad latitudinal regions (SI Appendix, Table S2). Fig. 4 is a two-dimensional representation of our MLR fits, where we have collapsed the multiple predictor concentrations into a single x variable. In this presentation, which is analogous to a traditional one-variable EC, the y -residuals are identical to those in the MLR, while the observation uncertainty is shown as the impact of measurement uncertainty on the predicted fluxes. Our revised flux estimates are the result of applying the inversion-based MLR fit coefficients to the equivalent ATom-observed partial-column annual-mean regional concentrations. Our refined uncertainty estimates are derived accounting for uncertainties in the observations and the MLR fits, and the prediction interval of the inversions (Materials and Methods).

Table 1 presents the results of our ECFC calculations for F_{SET} , F_T , and F_{NET} , as well as results from the full v10 MIP inversion ensemble and the subset of experiments excluding OG observations. Most notably, the new estimates reduce the uncertainties with respect to the full-ensemble by 46 to 56%, with the largest relative reduction for F_T . The new central flux estimates do not differ significantly from the previous ensemble means, either including or not including the OG inversions. However, comparing to individual inversions and the experiment means, the new estimates clearly favor the inversions that did not use OG observations (Fig. 4 and SI Appendix, Text). If we decide to exclude the inversions using OG observations (OG and LNLGOGIS) on the basis of their apparent biases shown here and against other metrics as reported by Byrne et al. (9), then the uncertainty reduction represented by our ECFC with respect to the non-OG subset is 17 to 28%, with the largest reduction for F_{NET} (Table 1). These numbers are based on MLR fits to the full ensemble, but are similar (19 to 25% uncertainty reductions) if we only fit the IS, LNLG, and LNLGIS inversion results.

Although the differences between our ECFC central estimates and the full v10 MIP ensemble means are modest, we highlight that robust uncertainty improvements in zonal carbon budgets have been historically difficult to achieve. For example, over more than a decade of observational and modeling advances, the $1\text{-}\sigma$ spread in NET total flux from RECCAP1 (13) ($\pm 0.51 \text{ PgC y}^{-1}$) reduced by only 0.12 PgC y^{-1} to that from the v10 MIP non-OG

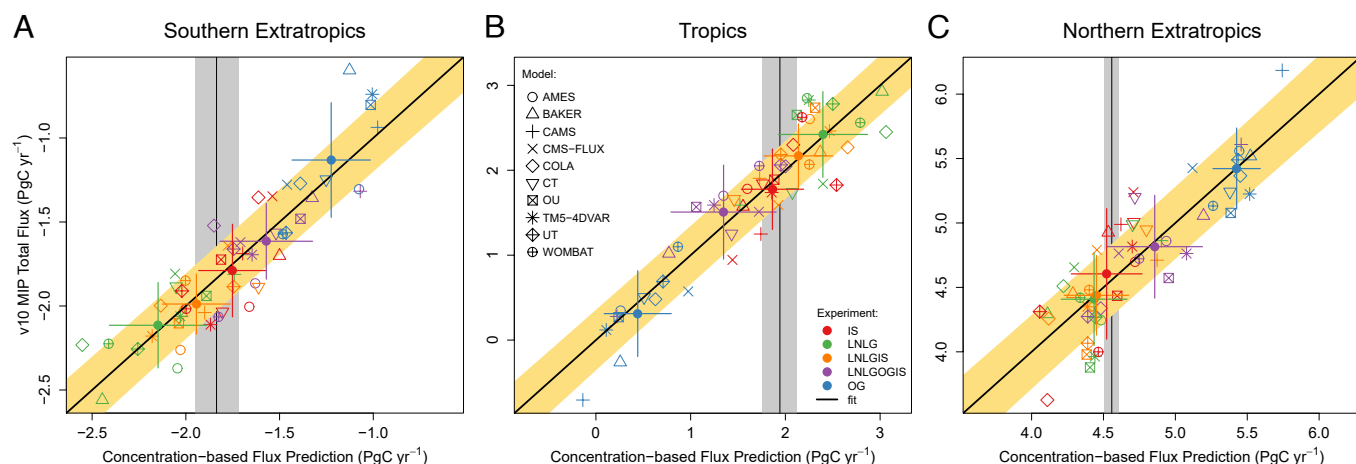


Fig. 4. Two-dimensional representations of the MLR fits between fluxes and concentrations from the v10 MIP for (A) southern extratropics, (B) tropics, and (C) northern extratropics. Each panel plots v10 MIP inversion fluxes averaged from June 2016–May 2018 (vertical axes) versus MLR predicted fluxes (horizontal axes) based on partial-column mean concentrations derived from ATOM cosamples for two or three regions as described in the text, with MLR coefficients in [SI Appendix, Table S2](#). Individual inversions are plotted as open symbols colored by experiment. Solid circles with whiskers show experiment means and SDs. Diagonal black lines show the 1:1 lines resulting from the MLR fits and the yellow shading shows the sum of the 1- σ CI of the MLR fits and the 1- σ prediction interval derived from the MLR fit residuals ([SI Appendix, Text](#)). The vertical black lines show the MLR predictions using ATOM observations, which correspond to our emergent concentration-flux constraint central flux estimates. The gray bands show 1- σ uncertainties in the observation-based MLR prediction derived by comparing measurements from five independent instruments on the aircraft ([SI Appendix, Text and Table S4](#)).

inversions ($\pm 0.39 \text{ PgC yr}^{-1}$). Thus, our estimate, which reduces the F_{NET} uncertainty relative to the v10 MIP non-OG inversions 0.11 PgC yr^{-1} further, represents a comparable advance.

Implied Land Fluxes. The latitudinal distribution of total CO_2 fluxes we report here includes a large southern extratropical sink, a large tropical source, and as expected from fossil-fuel emissions, a large northern extratropical source ([Table 1](#)). Of the three components of the total flux, fossil-fuel emissions and net air-sea flux have independent constraints, whereas uncertainties on independent net land flux estimates are much larger. To highlight the implications of our results for land CO_2 fluxes, we also show in [Table 1](#) pCO_2 -based air-sea CO_2 fluxes as compiled by the 2024 Global Carbon Budget (GCB 2024, [8](#)), canonical estimates of fossil-fuel emissions from national reporting as compiled by the GridFED emissions dataset ([37](#)), and the result of subtracting these from our ECFC total fluxes ([SI Appendix, Text](#)). The land fluxes implied by this subtraction ([Table 1](#)) suggest a large 2.2 PgC yr^{-1} northern extratropical sink, a 0.4 PgC yr^{-1} tropical source that is modest given estimates of large tropical land-use emissions (e.g., ref. [40](#)), and a 0.6 PgC yr^{-1} southern extratropical sink that is relatively large for the region's smaller land area.

[SI Appendix, Figs. S9 and S10](#) show these results, for total and land-only fluxes, in the context of previous inversion syntheses. Of these, the Stephens et al. ([33](#)) study is notable for using an early version of an ECFC approach that employed vertical CO_2 concentration gradients in the Northern Hemisphere to refine zonal land fluxes estimated by the TransCom 3 Level 2 study ([41](#)). The presence of large trends in total fluxes, for example $+3 \text{ PgC yr}^{-1}$ over 30 y in the northern extratropics, and large interannual variability in land fluxes, for example a range of 2 PgC yr^{-1} in the tropics, makes it difficult to compare results reported for different time periods. Nonetheless, the results from our study, using canonical fossil-fuel emissions, and previous syntheses are generally consistent with an increasing total source in the northern extratropics driven by fossil-fuel emissions, an increasing total sink in the southern extratropics driven by ocean uptake, and an increase in northern land sinks (e.g., ref. [28](#)).

Comparing the land sinks implied by canonical fossil and air-sea fluxes ([Table 1](#)) to in situ inversion estimates for the concurrent June 2016–May 2018 time frame from GCB 2024 ([8, 42](#)), which also used GridFED fossil-fuel emissions, we find general consistency with slightly lower land SET uptake, no T emission, and slightly lower NET uptake in GCB 2024 (-0.40 ,

Table 1. Zonal-mean fluxes (positive to the atmosphere) for June 2016–May 2018

	SET (> 20°S)	T (20°S–20°N)	NET (> 20°N)
v10 MIP total flux full ensemble	-1.73 ± 0.43	1.64 ± 0.88	4.74 ± 0.53
v10 MIP total flux no OG subset	-1.96 ± 0.27	2.12 ± 0.52	4.48 ± 0.40
ATOM + v10 MIP ECFC total flux	-1.84 ± 0.22	1.94 ± 0.38	4.56 ± 0.29
pCO_2 -based sea-to-air flux (36)	-1.66 ± 0.27	0.58 ± 0.20	-1.43 ± 0.23
Canonical fossil-fuel emission (37)	0.41 ± 0.03	0.98 ± 0.10	8.19 ± 0.55
Canonical implied land-to-air flux	-0.57 ± 0.36	0.36 ± 0.44	-2.17 ± 0.66
WLSH fossil-fuel emission (26)	0.39	0.92	7.70
WLSH implied land-to-air flux	-0.55	0.42	-1.68
Bottom-up land- to-air flux (8, 38, 39)	-0.17 ± 0.27	-0.70 ± 0.94	-1.41 ± 0.60

Fluxes are in units of PgC yr^{-1} . Uncertainties are 1- σ . The primary ECFC results of this study for total (land + ocean + fossil) and residual land fluxes are highlighted in boldface. The implied land-to-air fluxes are the result of subtracting the pCO_2 -based sea-to-air fluxes and either the canonical or weak land sink hypothesis (WLSH) fossil-fuel emissions from our ECFC total fluxes. For comparison purposes, the last row shows bottom-up land-to-air flux estimates based on dynamic global vegetation models (DGVMs) adjusted for land-use and land-use change (LULUC) and rivers.

0.02, and -2.01 PgC y^{-1} , respectively). Comparing to the Stephens et al. (33) land flux results for 1992–1996, we find similar estimates in the southern extratropics ($-0.62 \pm 0.97 \text{ PgC y}^{-1}$) and tropics ($0.13 \pm 0.78 \text{ PgC y}^{-1}$), but smaller northern extratropical uptake ($-1.53 \pm 0.64 \text{ PgC y}^{-1}$). The Stephens et al. (33) northern extratropical land sink represents 29% of the 1992–1996 fossil-fuel emissions used in that study, while our estimate for June 2016–May 2018 represents 27% of GridFED fossil-fuel emissions during this time. Our results and the Stephens et al. (33) study are consistent in revising both total and land fluxes up in the extratropics and down in the tropics relative to the corresponding TransCom 3 Level 2 and OCO-2 v10 MIP non-OG ensemble means (SI Appendix, Figs. S9 and S10).

Impact of Reduced Fossil Fuel Emission Estimates. Both GCB 2024 and our canonical estimates use the same fossil fuel emissions based on national reporting (37), and it is possible that these could be systematically biased high. Randerson et al. (26) argued for this possibility in presenting their weak land sink hypothesis (WLSH). This paper highlights a point also made by Fig. 14, namely that the global partitioning of land, ocean, and fossil carbon fluxes is still poorly constrained. Randerson et al. (26) start from satellite measurements of trends in above ground vegetation, which do not show large increases, and then explore the modest changes required in fossil and ocean estimates to match a weaker global land sink (0.8 PgC y^{-1} , not including uptake of C on land that is transported by rivers to the ocean). Their WLSH budget for 2000–2019 includes a 6% reduction in fossil fuel emissions and an increase in the GCB global ocean sink from 2.5 to 2.7 PgC y^{-1} (not including air-sea gas exchange of C transported by rivers).

Randerson et al. (26) prescribed these revised fossil and ocean fluxes, partitioned into the Northern and Southern Hemisphere, in a 2-box atmospheric inversion driven by surface observations from Mauna Loa and the South Pole, and found improved agreement with satellite-biomass-based northern hemisphere land sink estimates. Here, we use our airborne-based latitudinal partitioning of total CO_2 exchange along with WLSH-consistent fossil and ocean fluxes, to evaluate changes in our implied land fluxes, and compare both estimates to those from bottom-up process models (Table 1). To do this, we reduce the GridFED fossil-fuel emissions by 6% as shown in Table 1. The pCO_2 -based air-sea fluxes are already greater in magnitude than the GCB estimate, which averaged these with weaker uptake estimates from prognostic ocean models (8). From 2000 to 2019, these pCO_2 -based fluxes average -2.09 PgC y^{-1} , which combined with the global river flux estimate used by GCB equals -2.74 PgC y^{-1} . Because this is already consistent with the WLSH budget, we do not revise the air-sea fluxes further. Table 1 shows that the impact of reducing fossil fuel emissions is small south of 20°N , but leads to a large (0.49 PgC y^{-1}) reduction in the implied northern extratropical land sink. This revised northern extratropical land sink is much closer to that estimated by Stephens et al. (33), as shown in SI Appendix, Fig. S10, but would suggest a much smaller trend in northern land uptake than estimated by Ciais et al. (28). If the fossil-fuel emission overestimation started more recently than 1992–1996, as previously suggested for China by Saeki and Patra (43), then the weak land sink hypothesis would not require any modification to the Stephens et al. (33) result.

Comparisons to Bottom-Up Estimates. Comparisons between atmospheric inversions and terrestrial ecosystem process models

require consideration of CO_2 fluxes associated with land-use or land-use change (LULUC), and rivers, which have considerable uncertainty (e.g., ref. 44). Carbon transfers from land to ocean by rivers are not typically represented in DGVMs. LULUC fluxes can be represented in DGVMs by prescribing external land-use forcing, but in the framework of GCB they are treated separately. Starting with the 17 of 20 DGVMs included in GCB 2024 that provided monthly output (8), as modified by Friedlingstein et al. (38) and O’Sullivan et al. (39), we have added in LULUC and river fluxes (8, 38) to match what is seen by the atmosphere (SI Appendix, Text and Table S5) and show the resulting “bottom-up” air-to-land CO_2 fluxes in the last row of Table 1. In comparison to our results using canonical fossil fuel fluxes, the central DGVM estimates have much greater uptake in the tropics and much less uptake in the extratropics, with a tropical-minus-extratropical difference of 0.88 PgC y^{-1} versus 3.10 PgC y^{-1} in our result.

Modifying fossil fluxes according to WLSH brings the northern extratropical land flux estimate closer to agreement with the DGVM plus LULUC and river estimate. We conclude that the discrepancy between the land flux implied using canonical fossil estimates and the bottom-up estimates for the northern extratropics could be explained by an underestimation of uptake in the DGVMs or a combination of DGVM bias and overestimated fossil-fuel emissions. Uncertainties in LULUC or river fluxes could also play a role in explaining the northern extratropical discrepancies, but the magnitude of these adjustments (SI Appendix, Table S5) is relatively small. Underestimation of northern extratropical sinks in DGVMs could be a result of unresolved structural ecosystem changes; underestimation of regrowth fluxes (7); other nutrient, aerosol, or climate impacts on growth at high latitudes (5, 28); or in some cases possibly tuning to maintain a targeted global land sink. As described in Randerson (26), a host of socioeconomic drivers could lead to artificial inflation of fossil fuel combustion estimates. In the southern extratropics, owing to small fossil, LULUC, and river fluxes south of 20°S , the discrepancies are more likely to be tied to the DGVMs themselves.

In the tropics, reducing fossil fuel emissions does not help to improve the agreement between our airborne estimates and the bottom-up estimates, and actually makes the difference slightly larger. Here we conclude that biases in the bottom-up components, in particular the DGVM and LULUC estimates, are likely involved. We note that the bottom-up estimates shown in Table 1 have already been adjusted for the issues recently identified by Friedlingstein et al. (38) with DGVMs representing land-cover change and increases in C evasion from inland waters, and LULUC estimates including transient carbon stocks. We assign a fairly high uncertainty to tropical LULUC emissions of $\pm 0.62 \text{ PgC y}^{-1}$ (SI Appendix, Table S5). LULUC bookkeeping emission estimates are dependent on many factors with considerable uncertainty, including the spatial input datasets, assumed initial carbon densities, and modeled temporal response and recovery to change (8). An underestimate of tropical LULUC emissions could contribute to the strong bottom-up sink estimate. A more fundamental reason for the discrepancy between our airborne-constrained tropical land fluxes and the bottom-up estimates could be an overestimate of CO_2 fertilization in the models. Owing to the nature of photosynthesis parameterizations, and possibly also unrepresented nutrient feedback effects or microbial dynamics, CO_2 fertilization in DGVMs causes carbon uptake to scale strongly with net primary production (3), which may not be realistic.

Notably, Schimel et al. (3) used the results of Stephens et al. (33) and a combination of DGVM and inventory studies to argue for a strong fertilization effect. This conclusion was supported by general agreement on southern extratropical plus tropical land uptake derived from the same inversion models selected by Stephens et al. (33) ($-0.49 \pm 0.30 \text{ PgC y}^{-1}$) and five DGVMs with included CO_2 , climate, and LULUC forcing ($-0.70 \pm 1.40 \text{ PgC y}^{-1}$) for the 1992–1996 period. Combining southern extratropical and tropical fluxes with opposite signs and different drivers complicates the comparison to our results and its interpretation. Nonetheless, our summed southern extratropical and tropical land flux estimate for June 2016–May 2018, using canonical fossil-fuel emissions from national reporting, is $-0.21 \pm 0.57 \text{ PgC y}^{-1}$ while that for our bottom-up estimate is $-0.87 \pm 0.98 \text{ PgC y}^{-1}$, reflecting a much larger spread driven by changes in both the top-down and bottom-up flux estimates. We expect our flux estimates to be more robust, owing to our use of latitudinal gradients in column means rather than vertical gradients, with less sensitivity to atmospheric transport error (20), and our use of 50 inversions rather than 12, which leverage a greatly expanded observational network and several decades of model improvement. Better understanding the cause of the discrepancies between top-down atmospheric inversions, as refined here by global airborne surveys, and process-based bottom-up estimates represents a fundamental challenge to the global carbon cycle community.

Observing System Design. Interannual variability and trends necessitate robust observational constraints sustained for long periods of time, in order to characterize the response of the carbon cycle to short-term climate variability and long-term climate drivers. We conducted a similar analysis using latitudinal gradients derived from long-running surface stations, but found much worse correlations between fluxes and latitudinal gradients from these stations than from the airborne column means, resulting in larger flux uncertainties, though the central flux estimates were generally consistent with the results presented here. Challenges to using surface station measurements include local influences, representation error, and critical dependence on the accuracy of boundary-layer mixing in models. Thus, there is a strong need for global airborne surveys repeated regularly instead of only during episodic campaigns, to provide strong independent constraints on the large-scale partitioning of the global carbon cycle over time.

The advance of satellite greenhouse gas (GHG) measurements over the past decade has had a major impact on GHG monitoring, and the prospects for the future observing system. While satellites have already provided dramatic improvements in our ability to detect and quantify local-scale CO_2 and CH_4 emissions (i.e., power plant, landfill, feedlot, urban) (e.g., ref. 45), and the response of the carbon cycle to climate perturbations (e.g., ref. 46), their ability to quantify more diffuse time-average GHG exchanges (i.e., agricultural, ecosystem, wetland, ocean) is currently limited. Existing reflected solar instruments also lack measurements at high latitudes, particularly in winter, and in cloudy regions. It is clear that a future GHG observing system capable of providing reliable emission and uptake estimates at regional scales would rely heavily on satellites, but it must be anchored by in situ measurements, including vertical profiles through the atmospheric column, for calibration and validation of the satellite measurements, atmospheric transport constraints, and verification of the derived GHG emission and uptake estimates.

Subtracting fossil-fuel and ocean flux estimates from our ECFC total fluxes highlights the impact of uncertainty in different components of the carbon cycle on land flux uncertainties. Even using the new airborne constraint, the uncertainty on the NET land sink is greater than 25% and those on T and SET land fluxes are of comparable magnitude to the fluxes themselves. Furthermore, uncertainty in air-sea flux estimates still contribute to land flux uncertainty in inversions and global budgeting. Inversions that assume zero uncertainty on prescribed fossil-fuel emissions or apply tight prior uncertainties on ocean fluxes may give a false impression of greater land flux certainty. While global airborne surveys can reduce uncertainties in the latitudinal distribution of total fluxes, further clarity on the processes driving the current terrestrial carbon sink requires progress also be made on independent fossil-fuel and ocean flux estimates (26).

Improvements to the frequency, regularity, and global coverage of airborne CO_2 observations should play a role in achieving the necessary advances in flux estimation capabilities. NASA, NSF, NOAA, and DLR all now have long-range research jets that could be deployed to conduct repeated global airborne surveys of CO_2 and other greenhouse and related gases. If done collaboratively, such surveys of the troposphere would be analogous to the collaborative international program for hydrographic surveys of the ocean interior that has been conducted for decades (e.g. GEOSECS, JGOFS, WOCE, GO-SHIP). Private jets supported by nongovernmental funding could also be enlisted for this purpose. Furthermore, existing commercial aircraft observations could be expanded to provide many more profiles at much lower cost, albeit primarily over continental urban centers.

Summary and Future Perspectives

Our ECFC results reduce zonal-mean total CO_2 flux uncertainties by up to 56% with respect to the full v10 MIP ensemble and up to 28% with respect to the subset of inversions that did not use ocean glint satellite measurements. To obtain these results, we first identified coherent latitudinal patterns of atmospheric CO_2 variability across the inversions participating in the OCO-2 v10 MIP (*SI Appendix*, Fig. S3), and showed that these patterns correlate strongly with similar latitudinal patterns in the zonal total fluxes estimated by these inversions (Fig. 3 and *SI Appendix*, Fig. S3). We then fit inversion fluxes with inversion partial-column annual-mean concentrations for large regions (*SI Appendix*, Table S2) and used the resulting coefficients and observations from global airborne surveys to refine the flux estimates in an emergent concentration-flux constraint framework (Fig. 4). Subtracting independent estimates for fossil-fuel emissions and air-sea CO_2 fluxes from our ECFC total fluxes results in zonal mean residual land fluxes showing stronger extratropical uptake and weaker tropical uptake than predicted by bottom-up estimates (Table 1).

The discrepancies in the northern extratropics are most likely explained by either the DGVMs underestimating climate, nutrient, or growth driven enhancements to extratropical sinks, or by a combination of DGVM bias and systematic overestimation of fossil-fuel emissions in national reporting (26). In the tropics, the differences between the airborne-constrained and bottom-up estimates could be explained by the DGVMs overestimating CO_2 fertilization, or by a combination of DGVM bias and underestimated emissions in the LULUC products. In the southern extratropics, there is little room for fossil or LULUC errors, and the disagreement with our airborne estimates most likely reflects DGVM biases. Improvements to DGVM representation

of nutrient limitations, microbial dynamics, and evolving land cover; to LULUC bookkeeping land-use inputs, carbon densities prior to disturbance, and response curves following disturbance; and to validation of fossil-fuel emission estimates and inventories independent from national reporting, could all help to resolve these discrepancies. Comparisons between airborne-constrained latitudinal flux distributions and independent observation-based fluxes for the same zones, for example from trends in satellite biomass measurements as in Randerson et al. (26), would also be valuable. The resolution of these top-down versus bottom-up differences will have important implications for process understanding and predicting carbon-climate feedback effects.

No global-scale GHG surveys have been flown since ATom ended in 2018, but our results establish a benchmark for the expected benefits to latitudinal carbon flux partitioning if surveys with similar spatial and temporal coverage could fly in the future. For the ECFC approach here, the signal to noise, or ratio of inversion spread to observational uncertainty from June 2016–May 2018, is high enough that further uncertainty reductions beyond this benchmark must come from reducing the scatter in the flux versus concentration correlations (29). Much of this scatter is likely owing to transport biases, but further work is needed to separate the impact of transport errors from other effects such as differences in inversion methodology, or assigned prior uncertainties.

Reducing biases in satellite CO₂ measurements would also greatly help. Measuring CO₂ from space is exceedingly difficult, with relative precision demands for accurate regional to global fluxes of 0.1% or smaller. The MLR coefficients in *SI Appendix, Table S2* range in magnitude from 0.6 to 1.8 PgC y^{−1} per ppm, implying that latitudinal satellite biases of only a few tenths of ppm can result in large biases in the inferred latitudinal distribution of fluxes. The OCO science team has invested a great deal of effort for more than a decade to understand and reduce measurement biases (27, 47–49). If the remaining satellite CO₂ biases could be reduced further, the improvements to knowledge of global carbon budgeting could be transformative. However, OCO-2 is threatened by budget cuts, and there is a risk that future missions will focus on detecting strong local sources or extreme climate events, with less emphasis on the efforts to reduce large-scale biases that limit regional to global budgeting.

Further benefits from airborne surveys may be achievable if these observations could be ingested directly into the inversion models or data assimilation systems in a more intentional manner. For example, inversions could optimize to match airborne observations as partial-column means and possibly also as vertical gradients, subtracting lower and upper tropospheric values where the division is above the maximum boundary-layer height in order to negate model boundary-layer mixing biases. Furthermore, because airborne vertical gradients and column-mean latitudinal gradients collected with individual instruments or collections of instruments typically have very small observational uncertainty (*SI Appendix, Text, Table S3*) and less sensitivity to transport biases, inversions could weight these derived quantities more heavily.

There are other reasons for optimism regarding further improvements to inverse models. For one, the OCO-2 v11 MIP is currently underway, using a newer version of the OCO-2 dataset and updated inverse models. This version of the OCO-2 MIP is also including a set of forward simulations of common flux fields to more rigorously evaluate transport differences. Inversion methodologies are also continuing to advance, and computational bottlenecks are being pushed, in particular as high-performance computing hardware advances. In addition,

a new metric for defining large-scale meridional diabatic mixing rates from global airborne surveys has recently been developed and is being used to quantitatively evaluate atmospheric transport models (16, 20).

In order to understand and predict carbon-climate feedback effects, there is a strong need for a robust global system for flux estimation, leveraging satellite and in situ observations with advanced inversion or data assimilation techniques. We have demonstrated progress in determining the latitudinal distribution of total CO₂ exchange using the unique information provided by the ATom observations during 2016–2018. Moving forward, a collaborative international program of regular global airborne GHG surveys could be a vital component of the future flux estimation system.

Materials and Methods

ATom CO₂ Observations. The ATom Mission used the NASA DC-8 aircraft to conduct four global surveys at different times of year from July 2016 to May 2018 (21). Each survey included an Eastern Equatorial Pacific roundtrip flight from California, followed by a Pacific transect from north of Alaska to south of New Zealand, and then an Atlantic transect from Punta Arenas, Chile or Antarctica to Greenland and the Arctic. The flights spanned from 83°N to 86°S, with the plane continuously profiling vertically from near the ocean surface to the upper troposphere or lower stratosphere. The ATom payload included three in situ instruments measuring CO₂ continuously—a NOAA Picarro G2401-m (50), the Harvard Quantum Cascade Laser Spectrometer (QCLS, 51), and the NSF NCAR Airborne Oxygen Instrument (AO2, 52)—and two whole air samplers supporting CO₂ measurements—NOAA Portable Flask Packages (PPFs, 53) and the NSF NCAR / Scripps Medusa Flask Sampler (52). Here we use the NOAA Picarro measurements for our primary analyses, with the exception of the first two research flights of ATom-1 where they are not available and we use QCLS instead. This selection also matches the NOAA ObsPack product used for cosampling the v10 MIP inversions (54). We use comparisons to the other ATom CO₂ instruments to evaluate potential measurement biases and to estimate uncertainty in derived metrics as described in *SI Appendix, Text and Tables S3 and S4*.

We use all ATom research flights conducted primarily over ocean, excluding only the last two flights of ATom-1 over North America, with high CO₂ variability and inversion divergence [see Thompson et al. (21) for flight tracks]. We filter the ATom observations to exclude strong local influences when sampling boundary-layer air during take-offs, landings, low profiles over land, or missed-approaches at airports. The times at which this filter is applied are provided in the online code repository (Data Availability). This filter excludes 2.8% of the ATom CO₂ data. We also filter the observations to exclude stratospheric air using coincident measurements of N₂O (51), O₃ (55), and H₂O (56), whereby samples are considered stratospheric when H₂O is less than 50 ppm and either O₃ is greater than 150 ppb or N₂O detrended to a reference year of 2009 is less than 319 ppb (57). For ATom-1, where QCLS N₂O is missing, we use the Unmanned Aerial Systems Chromatograph for Atmospheric Trace Species (UCATS, Hurst, 2011). This filter excludes 12% of the ATom CO₂ data. We filter all cosampled inversion output identically to the observations. To combine results from different years, we detrend the observations and inversion output by subtracting a 25-mo smoothed and deseasonalized trend fit (58) to the observed monthly mean NOAA in situ record from Mauna Loa Observatory (MLO). *SI Appendix, Fig. S5* shows pressure-latitude cross-sections for each of the eight ATom transects—one Pacific and one Atlantic per deployment.

OCO-2 v10 MIP Posterior Concentrations. The v10 MIP (9) represents the third in a series of community inversion intercomparison projects conducted by the OCO Science Team, with the prior two using Version 7 (22) and Version 9 (14) of the OCO-2 product. For the v10 MIP, all models used common sets of in situ and satellite observations and prescribed a common fossil-fuel flux field. All other aspects of the inversions, including transport model, meteorology, resolution, assimilation methodology, prior land and ocean fluxes, and assigned land

and ocean prior uncertainties, were free to vary. Further details on the v10 MIP experiment can be found in Byrne et al. (9) and at the project website (https://gml.noaa.gov/ccgg/OCO2_v10mip). The publicly available v10 MIP output includes 14 models, of which 11 were analyzed by Byrne et al. (9). Here we show fluxes from 13 v10 MIP models in Fig. 1, excluding JHU which does not match observed atmospheric growth rates. We then only analyze concentrations from the subset of 10 models that provided posterior output cosampled along the ATom flight tracks for all five experiments. The ATom observations were not used in the v10 MIP inversions. Details for all models are listed in *SI Appendix, Table S1*.

Pressure-Latitude Binning and Seasonal Fits. We bin average the filtered and detrended ATom observations, or inversion cosamples, by 100 hPa, 10 degrees latitude, and research flight within the latitude range 70°S to 80°N. We then fit the averages for each bin with a single harmonic and a constant term representing the annual-mean offset from the MLO trend, as shown in *SI Appendix, Fig. S6*. The resulting annual-mean cross-section is shown in Fig. 2. *SI Appendix, Figs. S7 and S8* show annual-mean inversion-observation difference cross-sections for each model and experiment.

ECFC Uncertainty Estimates. We use a similar Monte Carlo approach to those of Long et al. (15), Gaubert et al. (35), and Loechli et al. (34), to propagate uncertainties. In our implementation, we combine five sets of partial-column annual-mean observations derived from the five ATom CO₂ instruments (*SI Appendix, Text*) with 200 sets of coefficients from random correlated sampling of the MLR fit errors (*SI Appendix, Table S2*), and then each of these with 200 *y*-errors from random sampling of the prediction interval. Our ECFC flux estimates are then calculated as the mean of the 40,000 estimates using the NOAA Picarro instrument, and the uncertainty as the SD of the 200,000 estimates using all sensors, and shown as vertical black lines with gray shading in Fig. 4. Although the uncertainties on the intercept terms in *SI Appendix, Table S2* are relatively large, they are anticorrelated with the slope uncertainties such that the impact of fit uncertainties on the flux estimates are very small.

Williamson and Sansom (31) showed that ECs incorporate assumptions about the relationship or exchangeability between the model ensemble and reality, that have not traditionally been made explicit in EC applications. We acknowledge that transport or external flux biases common to all v10 MIP inversions could change the intercept or slope of the concentration-flux relationships in the MLR fits. However, we estimate the impact of this additional source of uncertainty to be small (*SI Appendix, Text*), and do not incorporate this into our formal uncertainty estimates. Not allowing for this exchangeability uncertainty is consistent with the traditional inversion ensemble approach of reporting the population mean or median and SD (9, 12–14), and ensures comparability between our uncertainty estimates and those of previous studies.

Data, Materials, and Software Availability. The ATom CO₂ data, as sampled by the models, are available from the NOAA CO₂ GLOBALVIEWplus v6.1 ObsPack (54). Other CO₂ and ancillary measurements from ATom are available from ref. 59. Output from the v10 MIP experiment, including posterior fluxes and ATom ObsPack cosamples are provided at the project website (https://gml.noaa.gov/ccgg/OCO2_v10mip). We use v10 MIP output downloaded on Sep. 9, 2025, including the updated version 2 of the gridded OG fluxes. All analysis code necessary to produce the results presented here is available at <https://github.com/NCAR/latitudinal-airborne-constraints>. GCB 2024 DGVM flux products are available from <https://mdosullivan.github.io/GCB>. GCB 2024 pCO₂-based air-sea flux products are available from ref. 36. GCB 2024 GridFED fossil-fuel emissions are available from ref. 60. GCB 2024 gridded river flux estimates are available from ref. 42. NOAA global CO₂ and N₂O concentrations, and Mauna Loa CO₂ concentrations, are available from the NOAA GML website (<https://gml.noaa.gov>). Other data products not otherwise publicly available

are included in the GitHub code repository. All other data are included in the manuscript and/or *SI Appendix*.

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